

Homework 4

4.660.)

Note that $\|Ax\| = \|x\|$ is equivalent to $\|Ax\|^2 = \|x\|^2$, so we can use them both as base assumptions. We want to show that A 's columns are orthonormal and that $m \geq n$. We can write $A = [a_1 \ a_2 \ \dots \ a_n]$ where a_i is the i th column of A . First we'll show that A 's columns are normal. Let $x = e_i$ for some $i \in \{1, 2, \dots, n\}$. Then

$$\|Ax\| = \|x\| \Rightarrow \|Ae_i\| = \|a_i\| = \|e_i\| = 1$$

Thus, all of A 's columns are normal. Now we want to show that A 's columns are orthogonal. Assume for contradiction that A 's columns aren't orthogonal. That means there exists i and j such that $i \neq j$ and $a_i^T a_j \neq 0$. Let $x = e_i + e_j$. Then

$$\begin{aligned} \|Ax\|^2 &= \|x\|^2 \\ \Rightarrow \|A(e_i + e_j)\|^2 &= \|e_i + e_j\|^2 \\ \Rightarrow \|a_i + a_j\|^2 &= \|e_i\|^2 + \|e_j\|^2 \\ \Rightarrow \|a_i\|^2 + \|a_j\|^2 + 2a_i^T a_j &= 2 \\ \Rightarrow 2 + 2a_i^T a_j &= 2 \\ \Rightarrow a_i^T a_j &= 0 \end{aligned}$$

This is a contradiction, thus A 's columns are orthogonal. As they're normal as well, A has orthonormal columns. As A has orthonormal columns, we can't have $m < n$. If we did then the max rank of A would be m , meaning that A has at most m linearly independent columns with n columns total, so there would exist a linearly dependent column a_k such that

$$a_k = c_1 a_1 + c_2 a_2 + \dots + c_{k-1} a_{k-1} + 0a_k + c_{n+1} a_{k+1} + \dots + c_n a_n$$

where at least one of $c_1, c_2, \dots, c_{k-1}, c_{k+1}, \dots, c_n$ is nonzero. WLOG suppose $c_j, j \neq k$, is nonzero. This would imply $a_k^T a_j = c_j$ which contradicts that A has orthonormal columns. Therefore A has orthonormal columns and $m \geq n$.

7.1060.)

a.

Let $A = \frac{1}{\sqrt{n}}I_n$ and $b = \frac{1}{\sqrt{n}}f$. Then

$$\begin{aligned} \|Ag - b\|^2 &= \left\| \frac{1}{\sqrt{n}}g - \frac{1}{\sqrt{n}}f \right\|^2 = \left(\frac{g_1 - f_1}{\sqrt{n}} \right)^2 + \left(\frac{g_2 - f_2}{\sqrt{n}} \right)^2 + \dots + \left(\frac{g_n - f_n}{\sqrt{n}} \right)^2 \\ &= \sum_{i=1}^n \frac{(f_i - g_i)^2}{n} = d \end{aligned}$$

We'll do something similar for c . Let

$$P = \frac{1}{\sqrt{n-2}} \begin{bmatrix} n^2 & -2n^2 & n^2 & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & n^2 & -2n & n^2 & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & n^2 & -2n^2 & n^2 & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \dots & n^2 & -2n^2 & n^2 \end{bmatrix} \in \mathbb{R}^{(n-2) \times n}$$

Then

$$\begin{aligned} \|Pg\|^2 &= \left(\frac{n^2 g_1 - 2n^2 g_2 + n^2 g_3}{\sqrt{n-2}} \right)^2 + \dots + \left(\frac{n^2 g_{n-2} - 2n^2 g_{n-1} + n^2 g_n}{\sqrt{n-2}} \right)^2 = \sum_{i=2}^{n-1} \left(\frac{n^2 g_{i-1} - 2n^2 g_i + n^2 g_{i+1}}{\sqrt{n-2}} \right)^2 \\ &= \sum_{i=2}^{n-1} \frac{(n^2 g_{i-1} - 2n^2 g_i + n^2 g_{i+1})^2}{n-2} = \frac{1}{n-2} \sum_{i=2}^{n-1} \left(\frac{g_{i-1} - 2g_i + g_{i+1}}{\frac{1}{n^2}} \right)^2 = c \end{aligned}$$

Therefore, we want to minimize $\|Ag - b\|^2 + \mu\|Pg\|^2$. This is now in the form of a multi objective least squares problem which we know the answer to:

$$g = (A^\top A + \mu P^\top P)^{-1} A^\top b = \left(\frac{1}{n}I_n + \mu P^\top P\right)^{-1} \frac{1}{n}f$$

This method always works, regardless of f . The multi objective least squares problem is solvable on the condition $\begin{bmatrix} A \\ \sqrt{\mu}P \end{bmatrix}$ is full rank which for us is true because $A = \frac{1}{\sqrt{n}}I_n$, so every column is linearly independent. Now we just want to obtain the two extreme cases, when $\mu = 0$ and when $\mu \rightarrow \infty$. When $\mu = 0$ we have

$$g = \left(\frac{1}{n}I_n + 0P^\top P\right)^{-1} \frac{1}{n}f = \left(\frac{1}{n}I_n\right)^{-1} \frac{1}{n}f = nI_n \frac{1}{n}f = f$$

This makes sense, because in the $\mu = 0$ case, we only care about minimizing distance between g and f , and this is minimized when $g = f$. As $\mu \rightarrow \infty$ we care more and more about minimizing $\|Pg\|^2$. That means for this extreme we want $\|Pg\|^2 = 0$. Note that this is equivalent to $g_i = \frac{g_{i-1} + g_{i+1}}{2}$ for $i \in \{2, 3, \dots, n-1\}$ i.e. every point in g , except for the first and last, is the midpoint of the points immediately before and after it. In other words, g is a straight line, which makes sense as we want to minimize curvature. We still want to minimize $\|Ag - b\|^2 = \left\|\frac{1}{\sqrt{n}}g - \frac{1}{\sqrt{n}}f\right\|^2$ as well, so as $\mu \rightarrow \infty$, we want g to be a straight line solution that minimizes $\left\|\frac{1}{\sqrt{n}}g - \frac{1}{\sqrt{n}}f\right\|^2$ which makes g the line of best fit for f .

b.

This question was done using Julia, the code and plots can be seen below. No critical points appear on the tradeoff curve shown, but both ends of the curve approach an axis which represent $\mu \rightarrow 0$ for the horizontal axis and $\mu \rightarrow \infty$ for the vertical axis.

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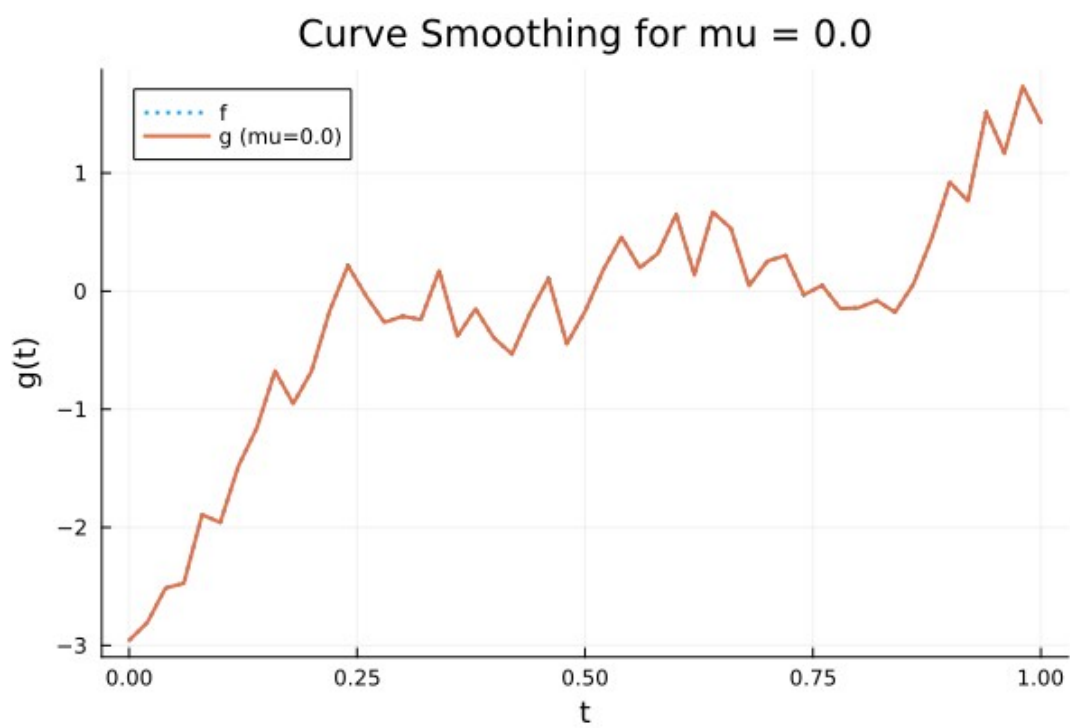
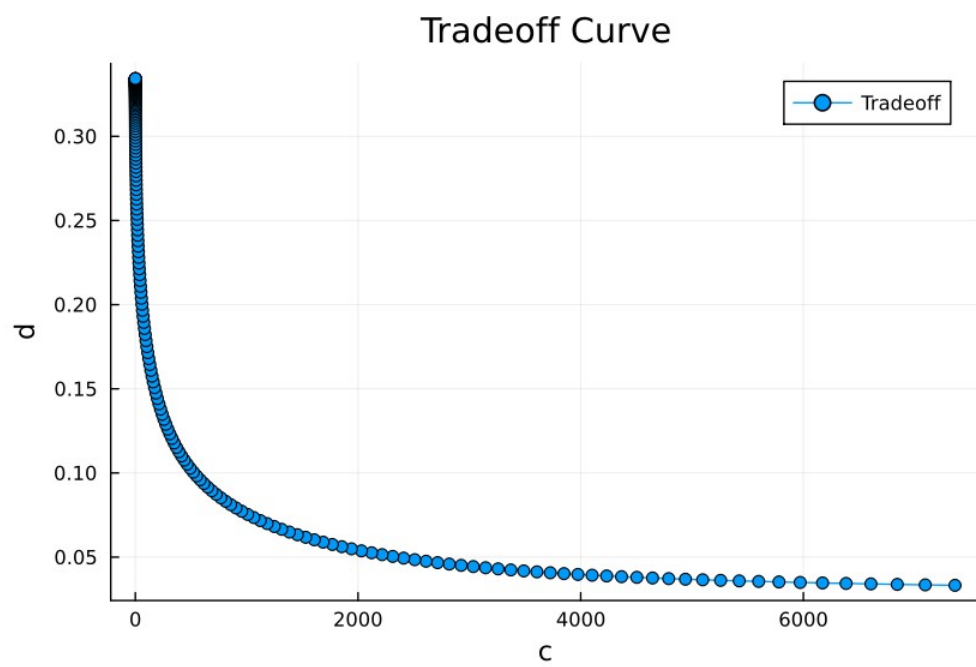
1  using LinearAlgebra
2  using Plots
3
4  include("readclassjson.jl")
5  data = readclassjson("curve_smoothing.json")
6
7  n = data["n"]
8  f = data["f"]
9
10 # Matrices
11 A = (1/sqrt(n)) * I
12 b = (1/sqrt(n)) * f
13 P = zeros(Float64, n-2, n)
14 for i in 1:n-2
15     P[i, i] = n^2
16     P[i, i+1] = -2*n^2
17     P[i, i+2] = n^2
18 end
19 P = (1/sqrt(n-2)) * P
20
21 # Tradeoff curve
22 mu_values = 10.^ range(-6, 0, length=200)
23 d = []
24 c = []
25
26 for mu in mu_values
27     g = inv(A' * A + mu * P' * P) * (A' * b)
28     push!(d, norm(A*g - b)^2)
29     push!(c, norm(P*g)^2)
30 end
31
32 p_tradeoff = plot(c, d, marker=:circle,
33                  xlabel="c",
34                  ylabel="d",
35                  title="Tradeoff Curve",
36                  label="Tradeoff")
37
38 # Smoothed curve
39 t = range(0, 1, length=n)
40 mu_values = [0.0, 1e-5, 1e-4, 1e-3, 1.0]
41 p_smooth = []
42

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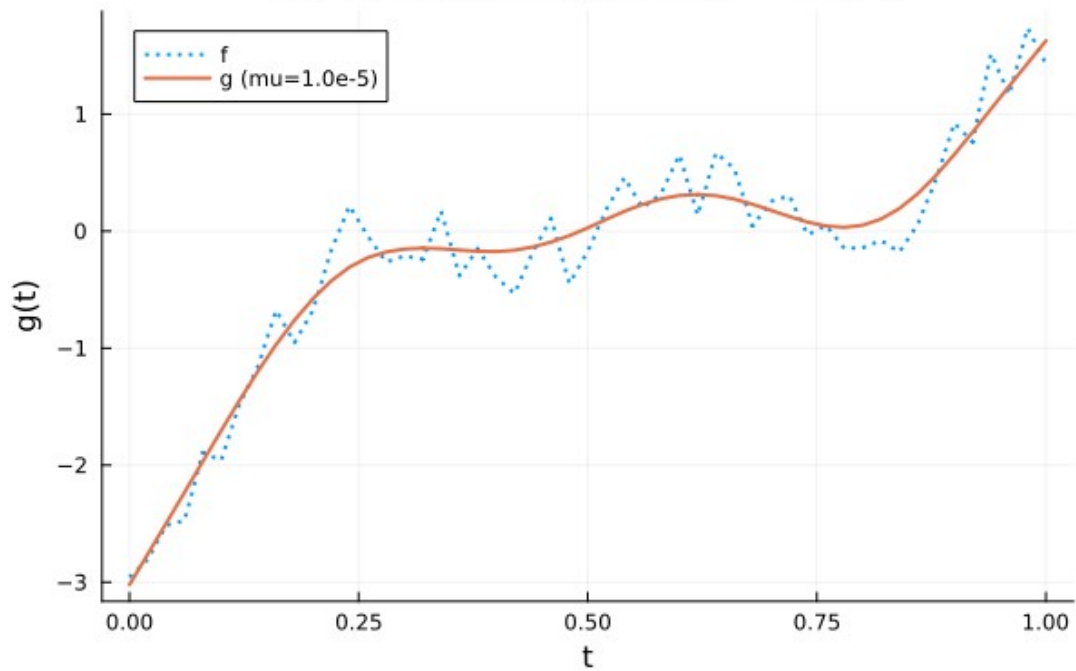
```

43 for mu in mu_values
44     g = inv(A' * A + mu * P' * P) * (A' * b)
45
46     p = plot(t, f, label="f", lw=2, linestyle=:dot)
47     plot!(p, t, g, label="g (mu=$mu)", lw=2)
48     xlabel!("t")
49     ylabel!("g(t)")
50     title!("Curve Smoothing for mu = $mu")
51
52     push!(p_smooth, p) # keep window alive
53 end
54
55 display(p_smooth[5])
56 println("Press Enter to exit...")
57 readline()
58

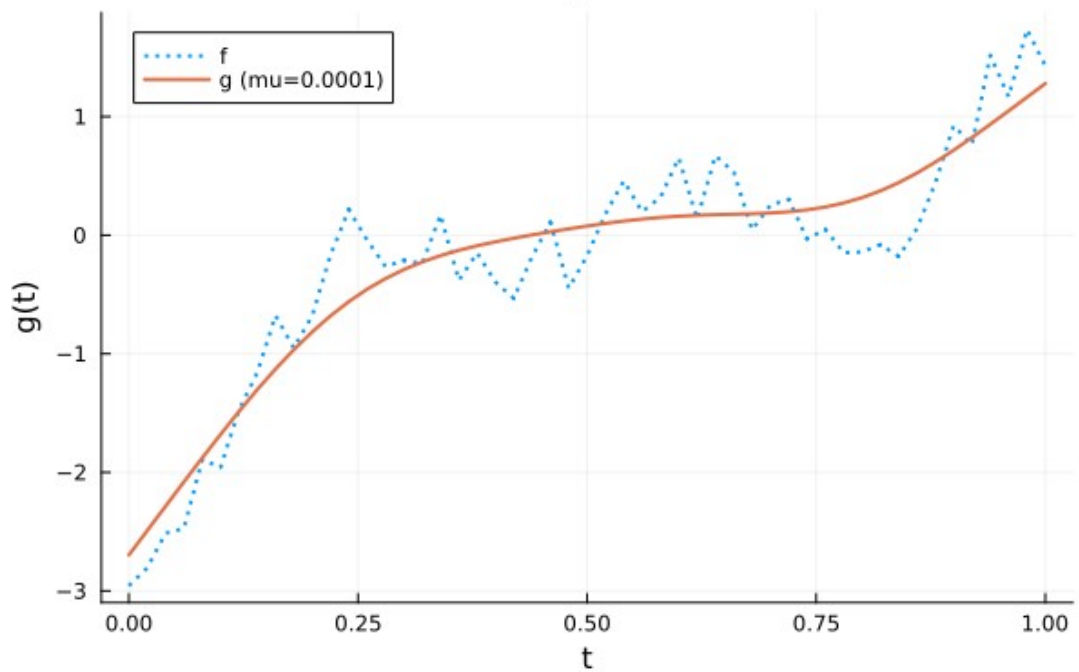
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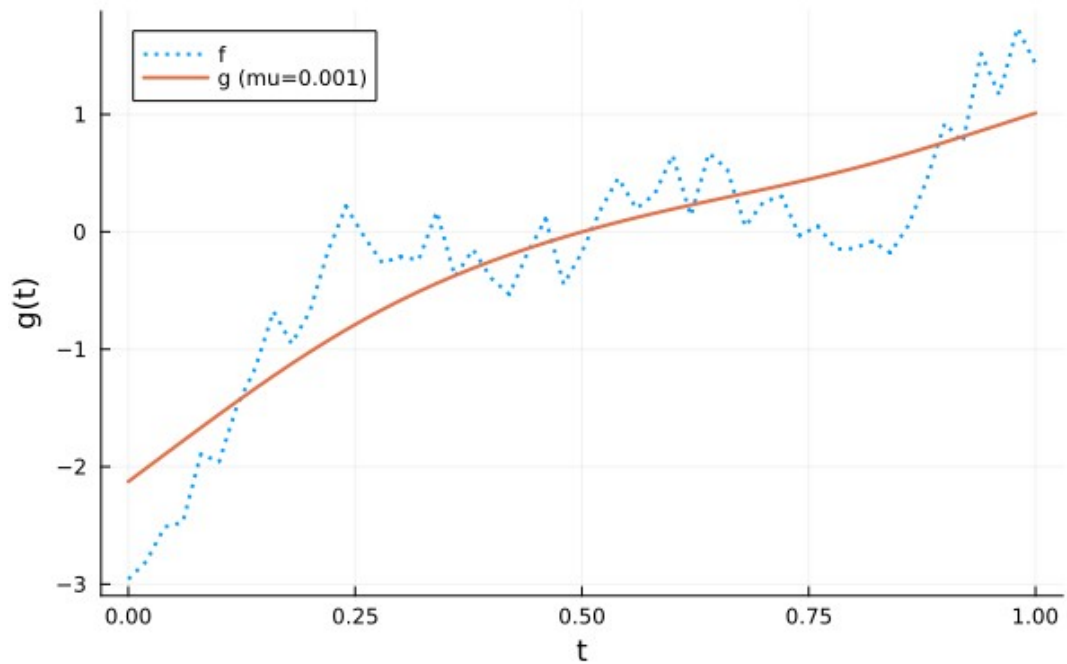
Curve Smoothing for $\mu = 1.0e-5$



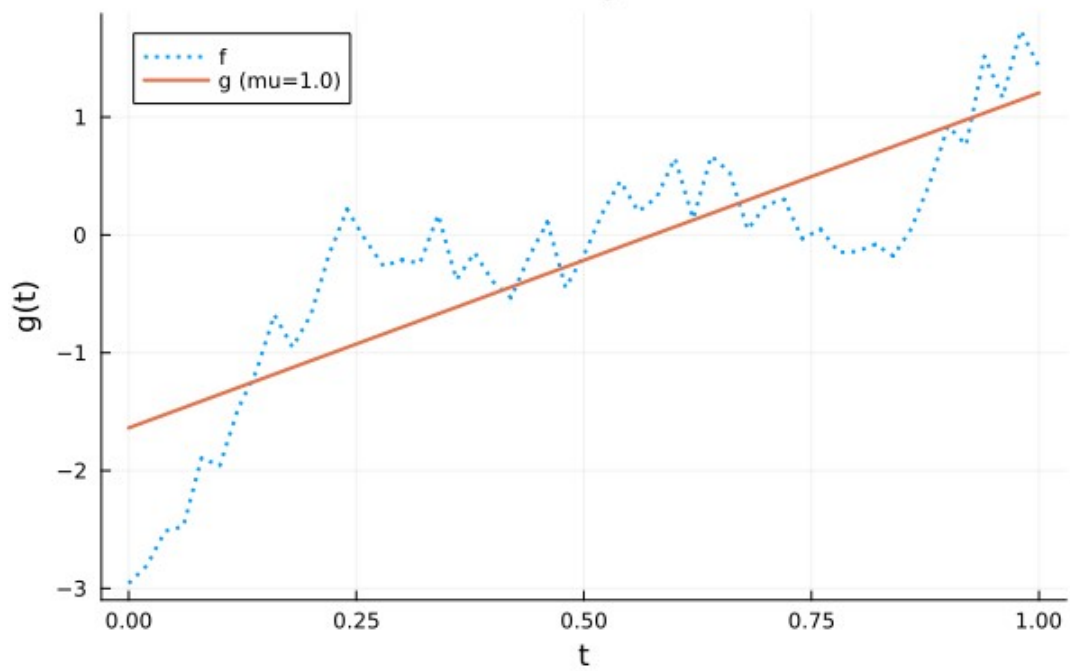
Curve Smoothing for $\mu = 0.0001$



Curve Smoothing for $\mu = 0.001$



Curve Smoothing for $\mu = 1.0$



8.1130.)

Let $A = \begin{bmatrix} a_1^\top \\ a_2^\top \\ \vdots \\ a_m^\top \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$. Then the conservation laws are equivalent to $Ay_{\text{adj}} = b$. Note that A has m rows and n columns where $m < n$, so A is fat. We want to find c such that $A(y + c) = b$ and $\|c\|$ is minimum. We can rewrite the equation to be

$$Ac = b - Ay$$

With this, we can give an explicit formula for the correction term

$$c = A^\top(AA^\top)^{-1}(b - Ay)$$

It can be seen that $(AA^\top)^{-1}$ is in our formula. This is not a problem because the rows of A are independent (as we assume that $a_1, \dots, a_m$ are independent) and A is fat, so A must be full rank. As A is fat and full rank, we have that $AA^\top$ is nonsingular. Now we just want to show that $y_{\text{adj}} = y + c$ satisfies the conservation laws and that c is the smallest possible correction term. We'll start by showing that $y_{\text{adj}} = y + c \Rightarrow Ay_{\text{adj}} = b$.

$$y_{\text{adj}} = y + c \Rightarrow Ay_{\text{adj}} = A(y + c) = Ay + Ac = Ay + AA^\top(AA^\top)^{-1}(b - Ay) = b$$

Therefore, y_{adj} satisfies the conservation laws. Suppose you had some other c' such that $A(y + c') = b$. Then we want to show that $\|c'\| \geq \|c\|$ or equivalently $\|c'\|^2 \geq \|c\|^2$.

$$\begin{aligned} A(y + c) &= A(y + c') \Rightarrow A(c' - c) = 0 \\ \Rightarrow (c' - c)^\top c &= (c' - c)^\top A^\top(AA^\top)^{-1}(b - Ay) = (A(c' - c))^\top(AA^\top)^{-1}(b - Ay) = 0 \Rightarrow c' - c \perp c \\ \Rightarrow \|c'\|^2 &= \|c' - c + c\|^2 = \|c' - c\|^2 + \|c\|^2 \geq \|c\|^2 \end{aligned}$$

Hence, c is the smallest possible correction term.